

8 How to read every graph

Every figure in the ICLR draft and the technical report, with the same four questions asked of each: what is on the axes, what one mark is, what to look for, and the sentence you should be able to say about it. Figures are shown small here; the full-size versions are in the documents. Read a graph in this order: axes first, then one mark, then the pattern, then the caption’s claim, and check that the pattern supports the claim.

8.1 The two figures in the ICLR paper

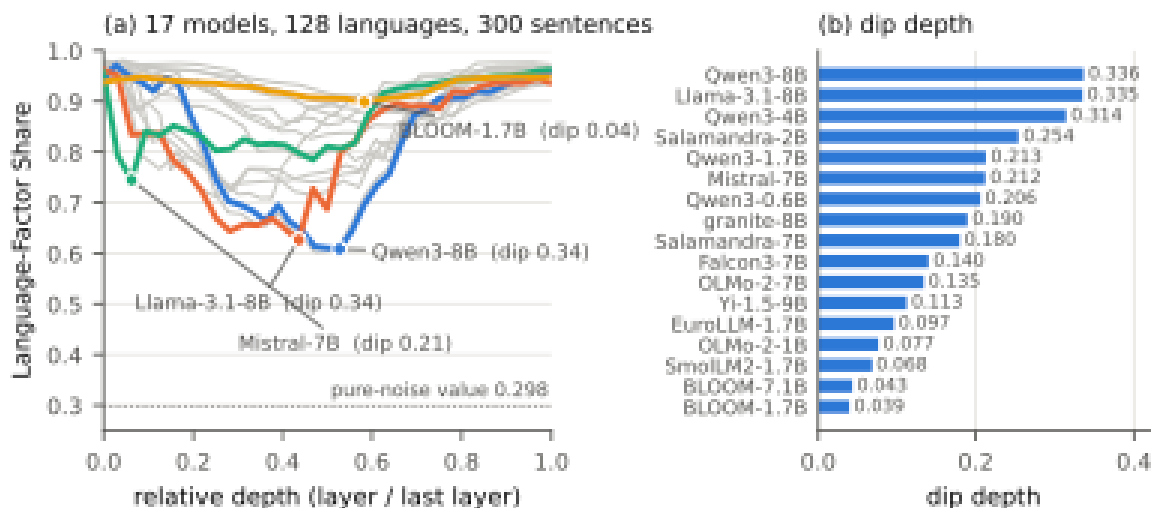


Figure 22: Paper Figure 1. (a) LFS against relative depth for 17 models; (b) dip depth per model.

Paper Figure 1(a). Horizontal axis: relative depth, layer index divided by the last layer, so that a 25-layer model and a 49-layer model can share one axis (0 is the embedding output, 1 is the last block). Vertical axis: LFS at that layer, computed on the 128 by 300 grid. One line is one model; 17 lines, four named. The dashed horizontal line at 0.298 is the pure-noise value $(L-1)/(L+N-2)$ for this grid size: a grid of random noise with no language or content structure would sit there. What to look for: every line starts high, near 0.93 to 0.98, drops to a minimum somewhere in the interior, and rises again toward the output; the depth of the drop differs enormously (Qwen3-8B and Llama-3.1-8B fall to about 0.61, BLOOM barely moves); every line stays above the noise value at every layer, so the language component is real everywhere, and the question is only how large it is relative to content. Sentence to say: “Representations start language-organized, become most meaning-weighted in the middle, and re-separate by language toward the output, in every model, by amounts that vary nine-fold.”

Paper Figure 1(b). Horizontal bars, one per model, sorted: dip depth, which is the first-layer LFS minus the minimum LFS. Read it as a ranking with the caveat from the bootstrap: neighbouring bars within about 0.02 are not distinguishable, and Mistral-7B and Salamandra-2B have wider intervals because their minimum sits at layer 2 or 6.

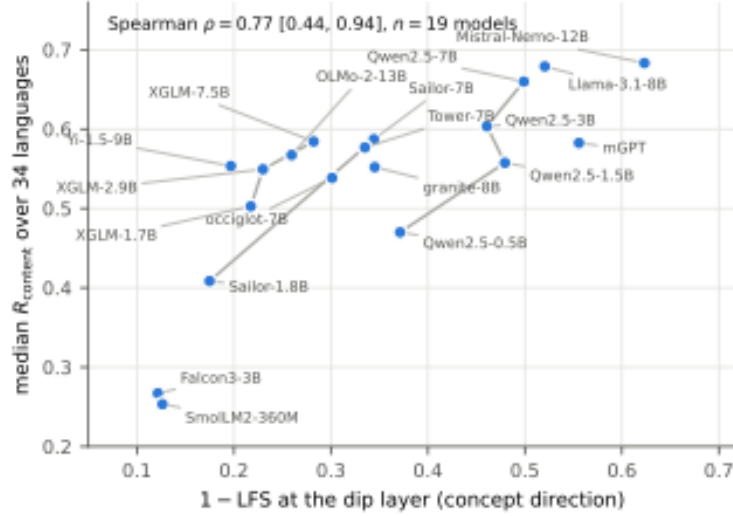


Figure 23: Paper Figure 2. The model-level content-transfer result.

Paper Figure 2. Horizontal axis: $1 - \text{LFS}$ at the dip layer, the concept direction, so models further right are more meaning-weighted at their dip. Vertical axis: the model’s median R_{content} over 34 languages, which is how much a foreign-language context sentence helps the model predict an English continuation when the context is the matched translation rather than a mismatched one, normalized by the English-context benefit. One dot is one of the 19 frozen models; gray connectors join sizes within a family. What to look for: the cloud rises to the right, Spearman 0.77 with model-bootstrap interval [0.44, 0.94]; SmolLM2-360M and Falcon3-3B sit bottom-left, Mistral-Nemo and Llama-3.1-8B top-right; within a family the connector also rises. Sentence to say: “Models whose middle layers weight meaning more make more use of foreign-language context, and the association holds at the level of whole models, not only pooled rows.” Caveat: 19 dots is a small sample; the interval is wide and the paper says so.

8.2 The technical report, in order

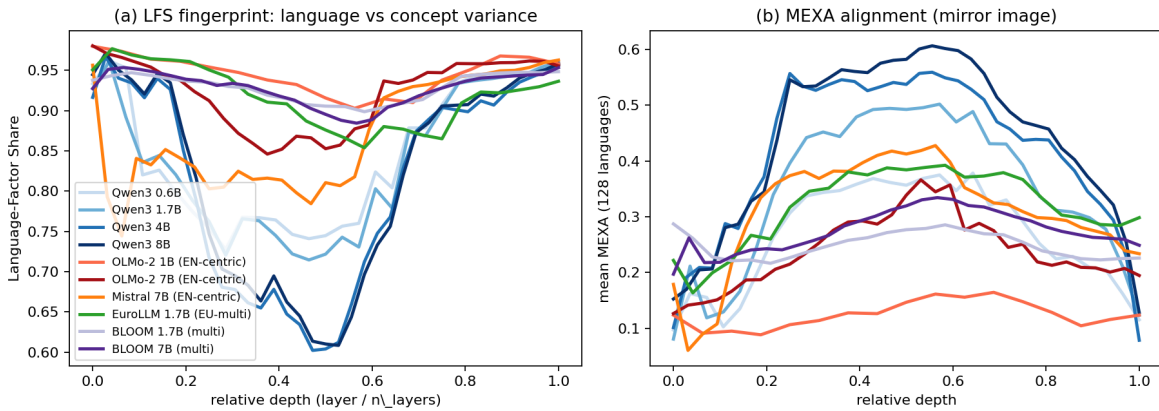


Figure 24: Report Figure 1. (a) LFS by depth, development grid; (b) mean MEXA by depth.

Report Figure 1. Same axes as paper Figure 1(a), for the ten development models, coloured by family (Qwen3 blues, OLMo-2 reds, Mistral orange, EuroLLM green, BLOOM purples). Panel (b) plots mean MEXA, the published retrieval measure, on the same depth axis. What to look for: (b) is roughly the mirror image of (a). Where the language share is lowest, retrieval of the matched translation is easiest. Sentence: “Two different instruments agree on where the middle is.” Do not read (b) as a competition; it is shown so a reader who knows MEXA can place LFS.

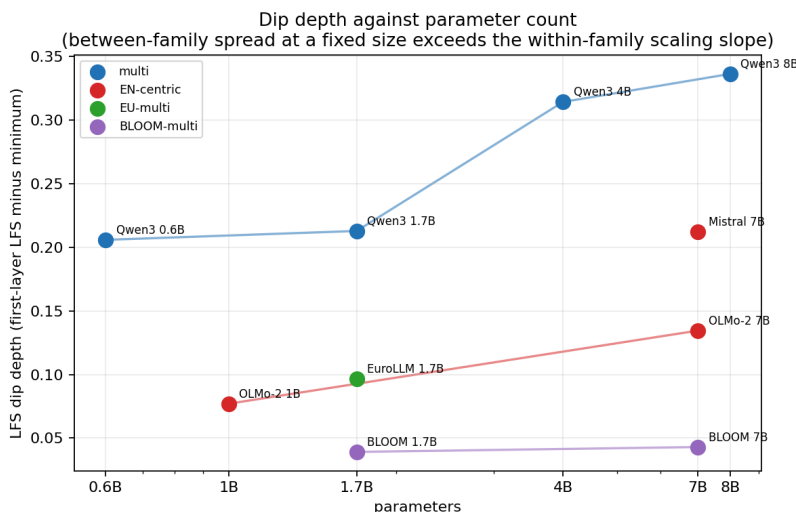


Figure 25: Report Figure 2. Dip depth against parameter count.

Report Figure 2 (dip depth against parameters). Horizontal axis: parameter count, on a compressed scale. Vertical axis: dip depth. One dot is one model, coloured by training-mix class; lines join sizes within a family. What to look for: within a family the line slopes up (scale deepens the dip), but at a fixed size the vertical spread across families is far larger than any slope. Near 7B the dots run from BLOOM at 0.04 through OLMo-2 at 0.13 and Mistral at 0.21 to Qwen3-8B at 0.34. Sentence: “How a model was trained moves the dip more than how big it is.” Caveat: dots within about 0.02 of each other, such as Qwen3-1.7B at 0.213 and Mistral-7B at 0.212, are not distinguishable once the bootstrap intervals are attached.

Report Figure 3. Panel (a) has two vertical axes: dip depth on the left (blue, solid) and low-resource-tier MEXA on the right (red, dashed), both against Qwen3 size. Read each line against its own axis only; the crossing point means nothing, because the two scales were chosen independently. What it shows: both rise with scale within Qwen3. Panel (b): for each model, three bars, MEXA at the best layer averaged over high-, mid-, and low-resource languages. What to look for: the low-resource bar is near the floor for OLMo-2, EuroLLM, and BLOOM, and rises with scale only for Qwen3. Sentence: “The training mix is visible tier by tier.” Caveat: this is a four-point trend for one family; the paper does not generalize it.

Report Figure 4 (heatmap). Rows are 127 languages sorted by their mean alignment across models; columns are the ten development models; colour is MEXA at the best layer, dark for 0 and yellow for 1. Panel (b) is a 26 language subset with labels and resource tiers. How to read:

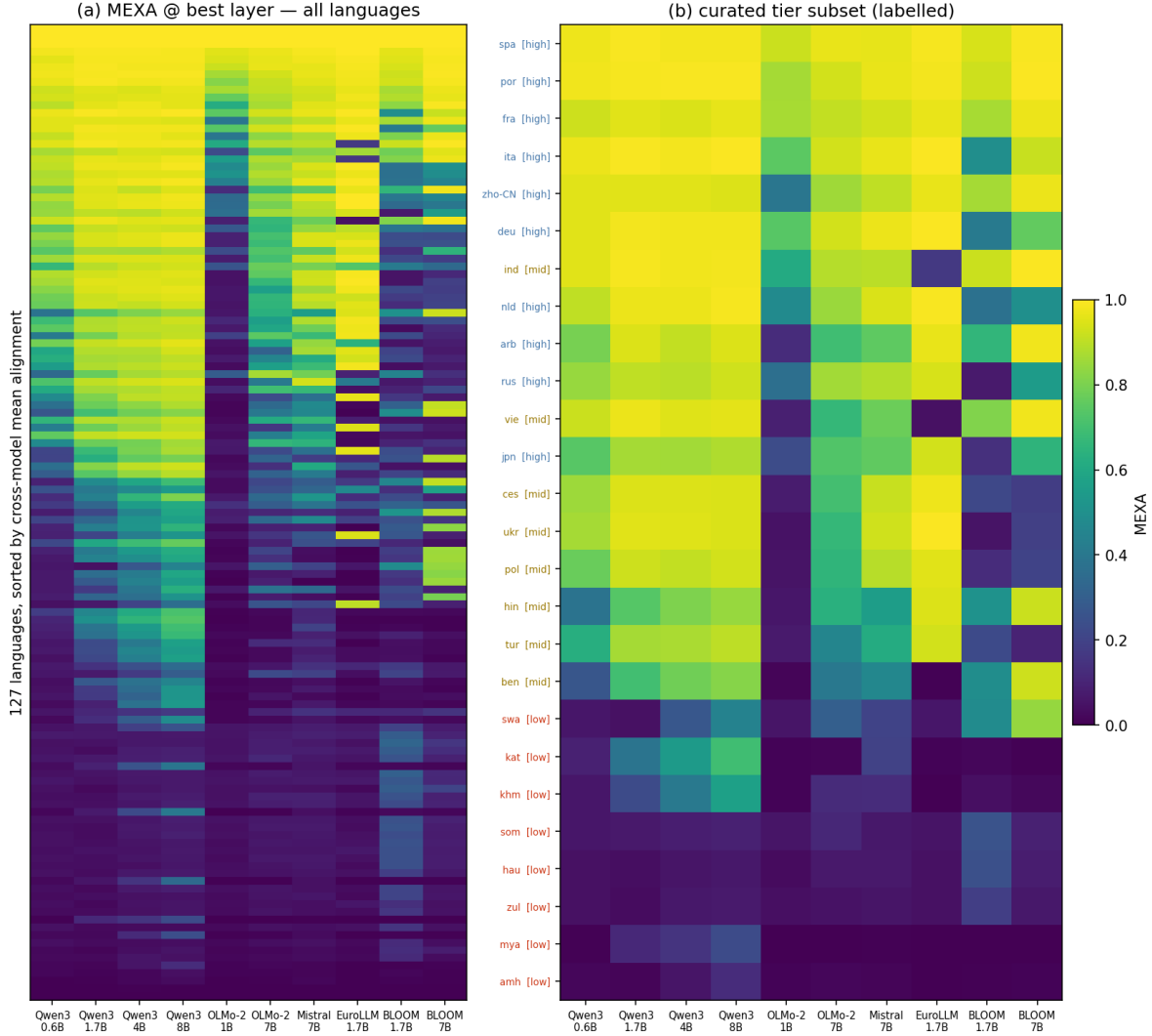


Figure 27: Report Figure 4. MEXA at the best layer, languages by models.

Report Figure 8 (Performance-at-Risk). One row per model; light bar is mean Belebele accuracy across 122 languages, red bar is PaR@10, the mean accuracy over the worst ten percent of languages. The dashed line is chance for a four-way exam, 0.25. What to look for: the red bar lags the light bar for every model, and for several models the red bar sits at chance while the mean is comfortably above it. Sentence: “The mean hides languages on which the model is guessing.”

Report Figure 9. The same two numbers as Figure 8, now as a scatter: mean accuracy horizontally, worst-decile accuracy vertically, dashed chance lines on both. What to look for: Qwen3-8B is alone in the top right, a cluster of models sits at the chance corner, and no model is high on one axis and low on the other. Sentence: “On this exam, serving the tail and serving the average go together; nobody buys tail robustness by sacrificing the mean.”

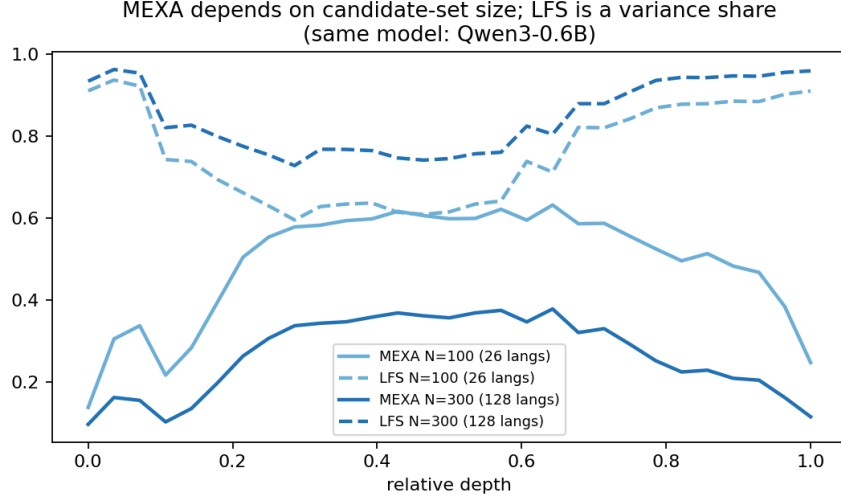


Figure 28: Report Figure 5. Evaluation-size dependence.

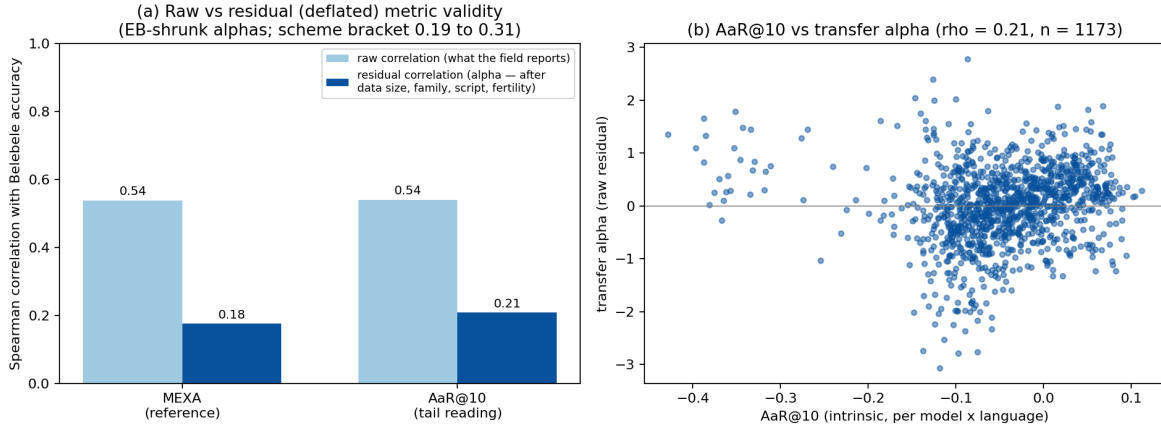


Figure 29: Report Figure 6. Raw and deflated validity against exam accuracy.

Report Figure 10. Panel (a), one model (Qwen3-4B): each dot is a language, horizontal is its alignment at the best layer, vertical is its content-specific context benefit R_{content} ; Amharic, Burmese, and Swahili sit bottom-left, German, Chinese, and Vietnamese top-right, Spearman 0.87. Panel (b): both axes replaced by their residuals after the same four controls, pooled over four models and coloured by model; Spearman 0.67. What to look for: the cloud in (b) still rises. Sentence: “The relation between alignment and context use is not just resource level in disguise.” This is the four-model pilot; the 19-model version is paper Figure 2.

Report Figure 11 (two events). Two vertical axes again, so read each curve against its own. Red, left axis: the share of probability mass the logit lens puts on tokens in the input’s script, which falls steeply by relative depth 0.2. Blue, right axis: LFS, which reaches its minimum near 0.47. Dotted verticals mark the two minima. Sentence: “The surface script is abandoned early; the language share of the variance converges later. They are two events, not one.” This is the report’s

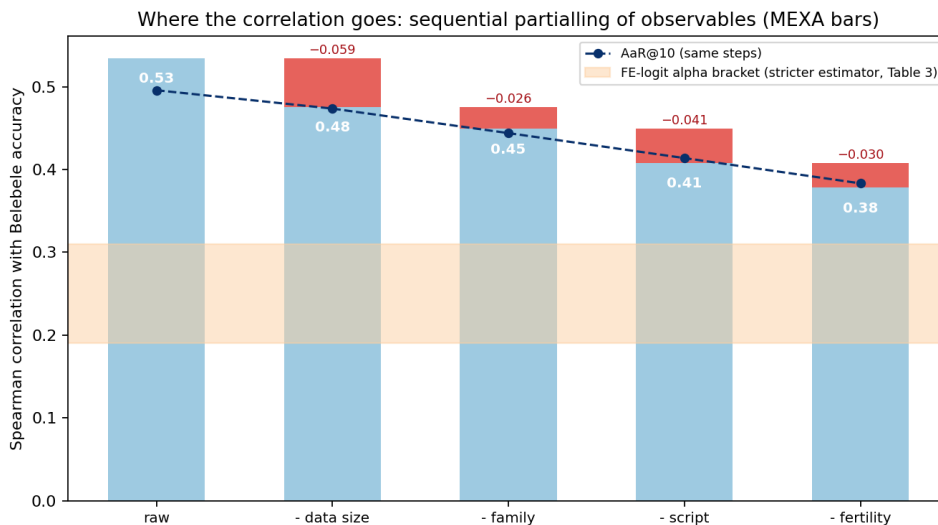


Figure 30: Report Figure 7. Sequential partialling of observables.

evidence that a logit-lens reading and a representation reading measure different things.

Report Figure 12 (misalignment budget). One stacked bar per model, at the dip layer, on the 1,500-sentence campaign data. The bar is the held-out cross-language misalignment, split into what a nested ladder of maps explains: an additive offset (dark), an isotropic scale (light), a general linear map (purple), a rotation (red, barely visible), and an unattributed remainder (hatched). Rotation shares are printed at the right. What to look for: offset plus scale explain 0.63 to 0.97 of the misalignment in every model, rotation never exceeds 0.7 percent. Sentence: “Languages differ mostly by a shift and a stretch, which is exactly what an additive decomposition can see; the construct boundary of LFS is empirically harmless here.”

Report Figure 13 (tail curves). Horizontal axis: which tail, from the worst 20 percent of sentences per language on the left to the worst 1 percent on the right. Vertical axis: the mean retrieval margin of that tail; negative means the true translation loses to a decoy. Gray lines are 13 models, four highlighted. What to look for: every curve descends to the right, so the worst sentences fail everywhere; the slope differs. OLMo-2-7B drops steeply toward the worst 1 percent (rare but severe failures), Qwen3-8B stays highest, BLOOM and SmolLM2 are flatter and uniformly weak. Sentence: “Two models with similar means can have different tails; the tail curve separates catastrophic from broad fragility.”

Report Figure 14 (ratio against components). Five training arms of Qwen3-0.6B on the word-alignment objectives. Left: LFS for each arm, all between 0.445 and 0.493. Right, logarithmic axis: raw concept variance at the same layer, 13,791 for the frozen model and 699 for the word-MSE arm. What to look for: the arm that lost 95 percent of its concept variance is indistinguishable from the sound arms on the left and stands out by a factor of twenty on the right. Sentence: “The ratio divides out the quantity that failed; report the components.” This is the single most important figure for the “why not a score” argument.

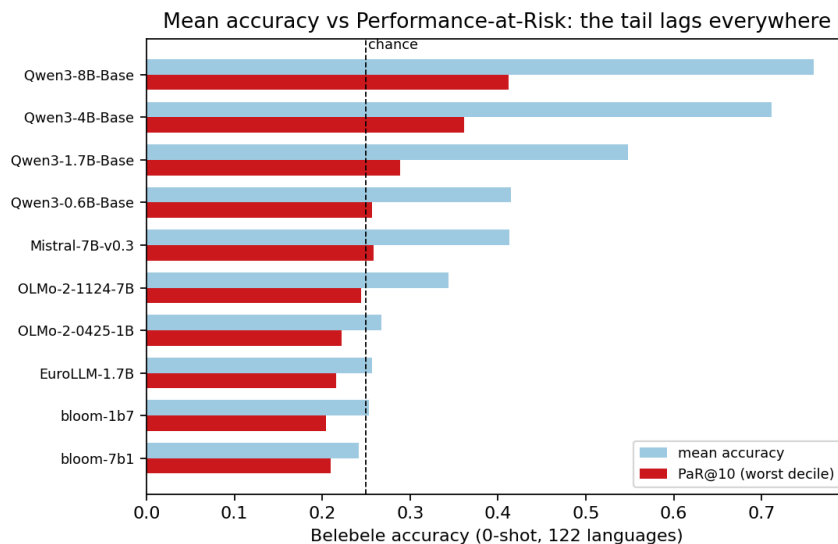


Figure 31: Report Figure 8. Mean accuracy against worst-decile accuracy.

Report Figure 15 (Aim 2 arms). Three panels for three objective families; within each, one group of three bars per arm: MEXA (retrieval), the magnitude of AaR (tail, lower is better), and CVP, concept-variance preservation relative to the frozen model. Dotted lines mark the frozen model’s retrieval and CVP of 1. What to look for: in the word-level panel, retrieval rises for the word-MSE and language-vector arms while their CVP falls to about 0.05; the cosine arm keeps CVP at 1 and gains nothing. Sentence: “Objectives that improve alignment scores can do so by shrinking the representation; only the preservation reading catches it.”

Report Figure 16. Horizontal axis: the code-switch rate used in training. Vertical axis: perplexity relative to the frozen model, so 1.0 is unchanged and higher is worse. Three groups of languages: those present in the switched data, those protected by replay, and those genuinely unseen. What to look for: the unseen group degrades monotonically, to 1.89 times worse at a 50 percent rate, while the other two groups sit at or below 1. Sentence: “Replay protects what it contains and nothing else.”

Report Figure 17. Horizontal axis, logarithmic: the weight on the concept-variance hinge penalty, from 1 to 1000. Vertical axis: mean concept-variance preservation. The blue curve is the sentence-level contrastive arm, rising from 0.91 to 1.02 and flattening. The dashed red line at 0.026 is the word-level arm with the same guard active. Sentence: “A bounded penalty saturates; it cannot restrain an unbounded alignment term, so the guard has to be a constraint, not a weight.”

Report Figure 18 (probe). Left: layers of Qwen3-0.6B on the horizontal axis; LFS in red on the left axis; accuracy of a classifier that predicts the language from the hidden state in blue (linear) and green (small network) on the right axis. Read the right axis carefully: it runs from 0.870 to 0.890, a zoom; chance is 0.042. Dashed verticals mark the LFS minimum at layer 13 and the probe minimum at layer 14. Right: the same 29 layers as a scatter, LFS against probe accuracy, Pearson 0.61. What to look for: the curves move together in direction, but the probe never drops

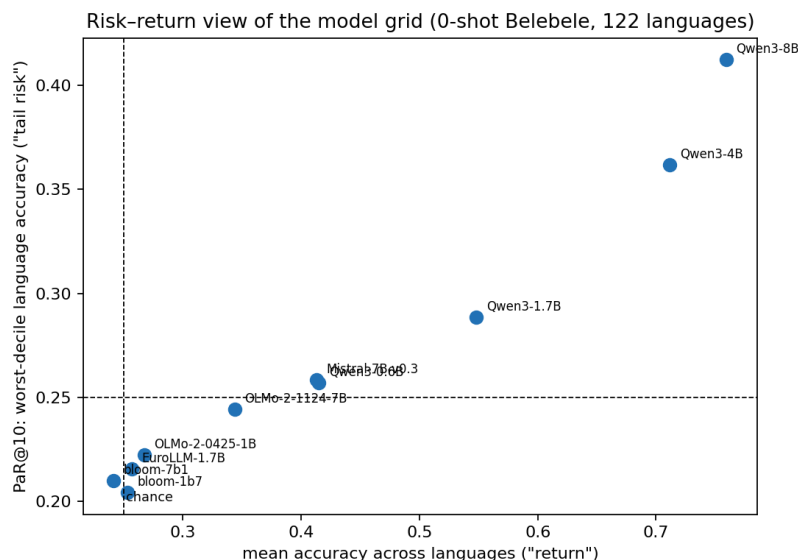


Figure 32: Report Figure 9. Mean accuracy against tail accuracy, one dot per model.

below 87 percent. Sentence: “Even at the dip, the language is almost perfectly readable from the representation; the dip is about the share of variance, not about erasing language identity.” This is why “lower LFS” must never be read as “language-agnostic”.

8.3 Reading habits worth keeping

- Find the null or the chance line before judging any level. LFS has 0.298 on this grid; the exam has 0.25.
- A depth axis is relative unless it says “layer”; the same relative depth is a different layer in a 25-layer and a 49-layer model.
- Two vertical axes (report Figures 3a and 11) invite a false reading of the crossing point. Read each curve against its own axis.
- A ratio next to a raw component (report Figure 14) is the habit the paper asks the field to adopt.
- Every number in a figure traces to an artifact in the claims ledger; if a reviewer asks where a bar came from, the answer is a file path and a job identifier.

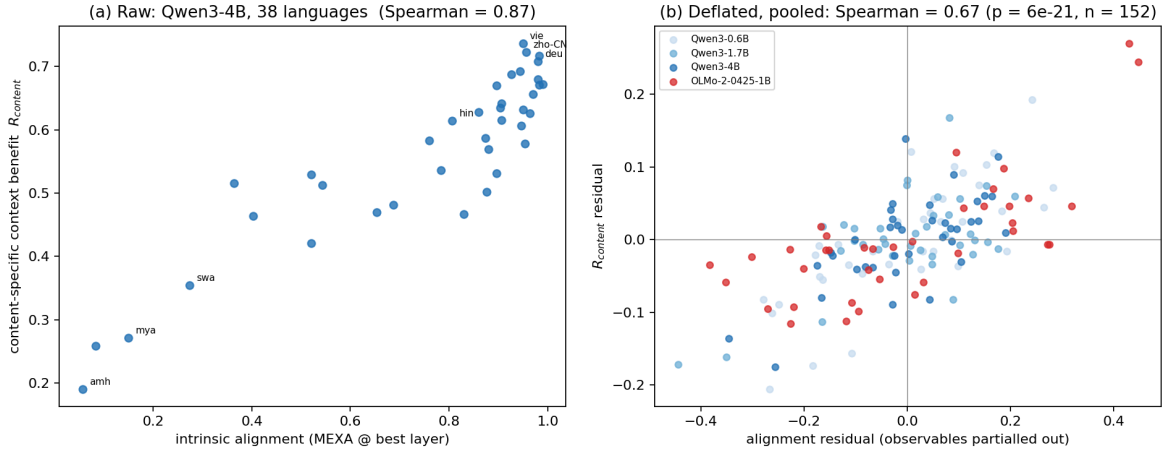


Figure 33: Report Figure 10. Alignment against content-specific context benefit.

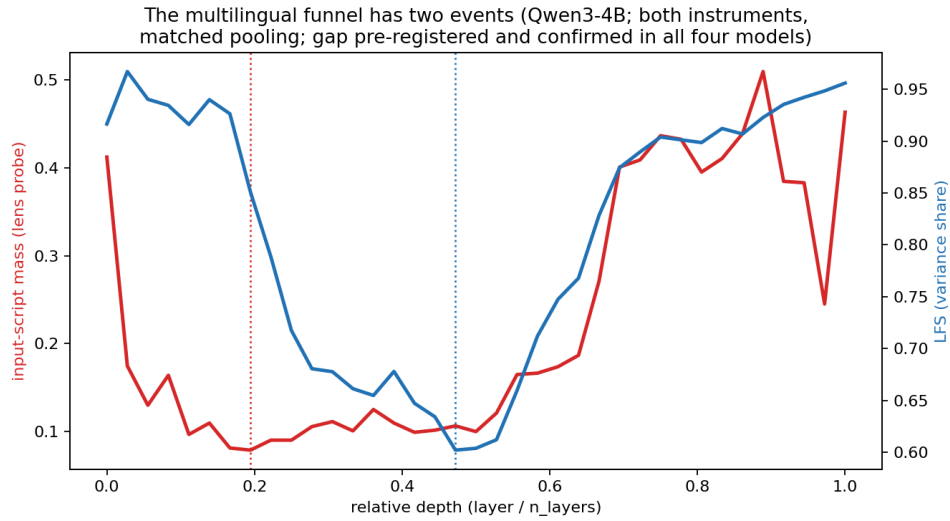


Figure 34: Report Figure 11. Two instruments on one depth axis.

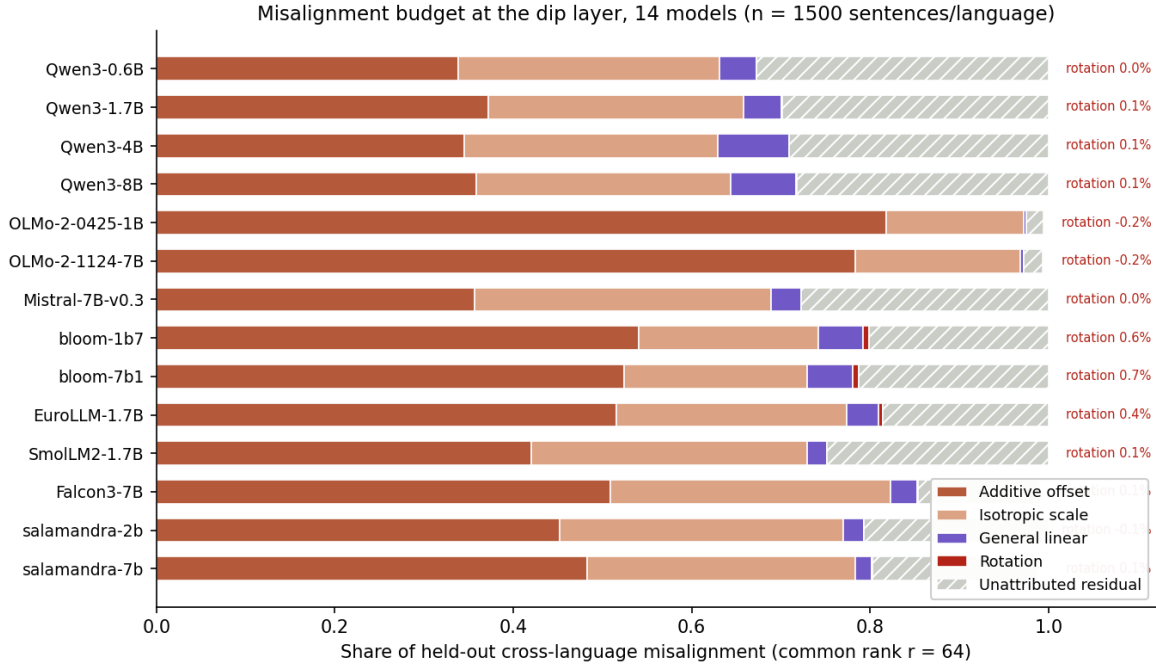


Figure 35: Report Figure 12 and paper appendix figure. The misalignment budget.

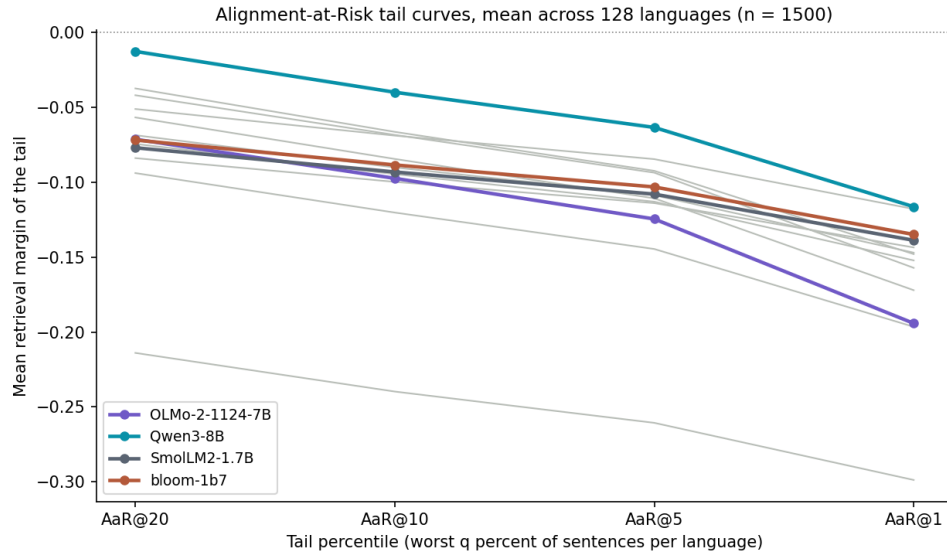


Figure 36: Report Figure 13. Alignment-at-Risk tail curves.

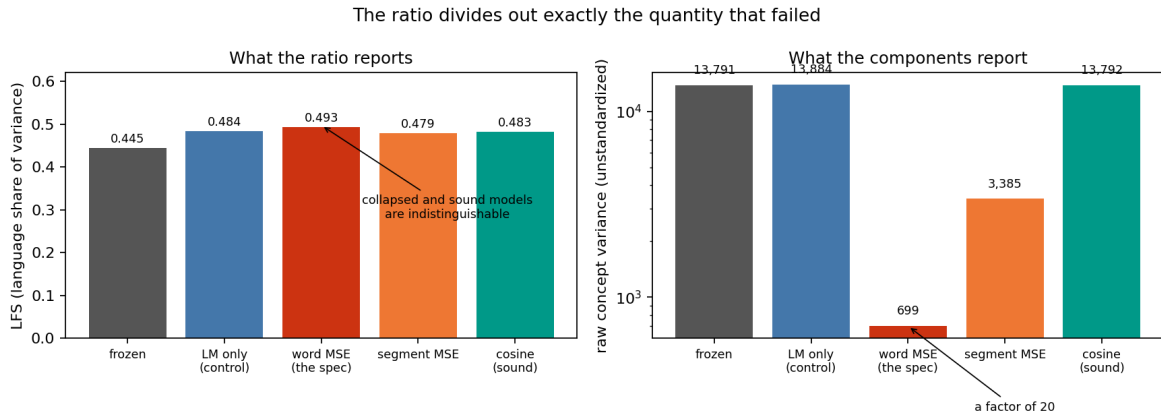


Figure 37: Report Figure 14 and paper appendix figure. The collapse arm read two ways.

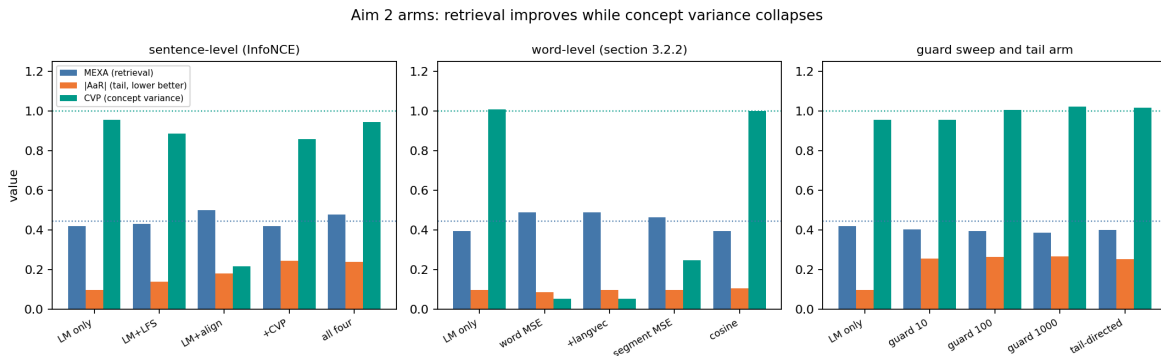


Figure 38: Report Figure 15. Every Aim 2 arm on three instruments.

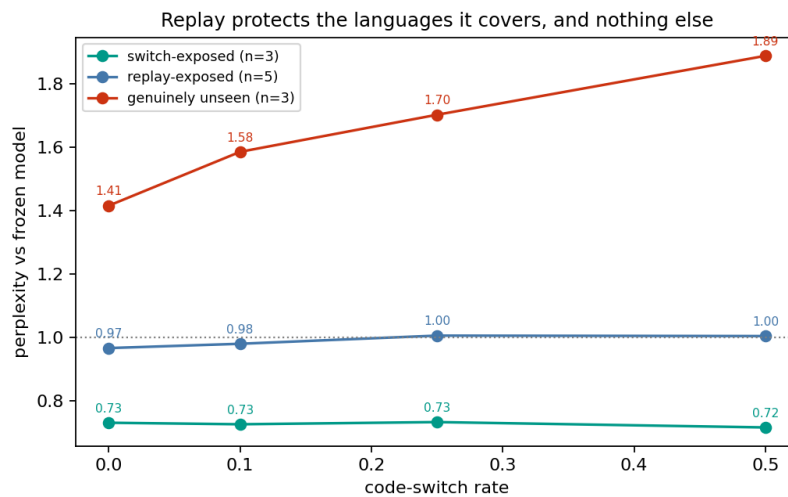


Figure 39: Report Figure 16. Code-switched training cost by language group.

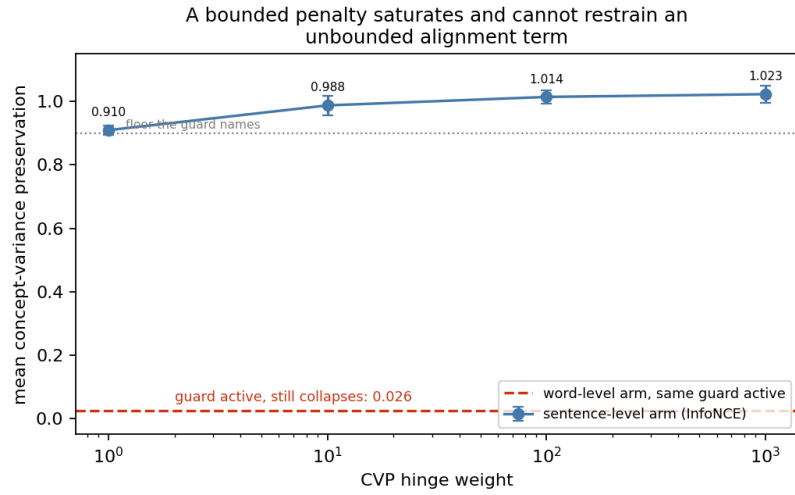


Figure 40: Report Figure 17. The concept-variance guard sweep.

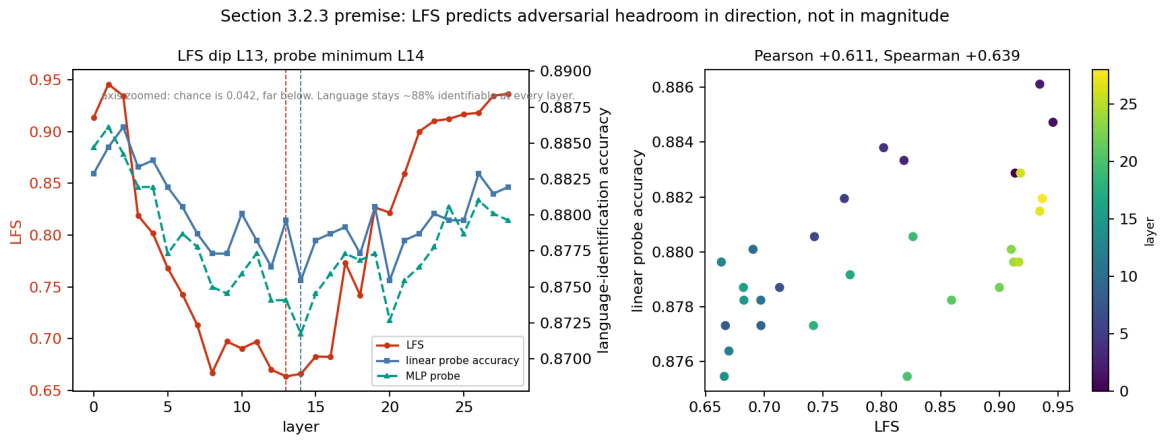


Figure 41: Report Figure 18. LFS against language-identification probe accuracy.